



The ProbLemma Channel Products

This document links all the products evaluated on the ProbLemma channel to the episodes in which these evaluations were carried out.



In addition, all the products related episodes are collected in the “Products Are Us” play list.

1. Season 1 Episode 9 (**S1M10**): One Cotangent, Two Cotangents (Transform And Conquer)

$$\prod_{k=1}^m \cot^2 \left(\frac{k\pi}{2m+1} \right) = \frac{1}{2m+1}$$

2. Season 2 Episode 41 (**S2M43**): 5-Coprime Odd Numbers That Live On A Globe

$$C_{40} = \prod_{a_i} \cos \left(\frac{a_i \pi}{100} \right) = \cos \left(\frac{\pi}{100} \right) \cdot \cos \left(\frac{3\pi}{100} \right) \cdot \cos \left(\frac{7\pi}{100} \right) \cdot \dots \cdot \cos \left(\frac{99\pi}{100} \right) = \frac{1}{2^{40}}$$

where a_i are the 5-coprime odd integers between 1 and 100:

$$\{a_i\} = \{1, 3, 7, 9, 11, 13, 17, 19, 21, \dots, 93, 97, 99\}$$

3. Season 2 Episode 43 (**S2M45**): A Poisson Integral By The Book (Divide And Conquer)

$$\prod_{k=1}^{n-1} \left(z^2 - 2z \cos \left(\frac{k\pi}{n} \right) + 1 \right) = \frac{z^{2n} - 1}{z^2 - 1}, \quad z \in \mathbb{C}$$

4. Season 3 Episode 36 (**S3M20**): Sine As A Product, With Applications

$$\sin(x) = x \cdot \prod_{n=1}^{+\infty} \left(1 - \frac{x^2}{n^2 \pi^2} \right)$$

for all real $x \neq \pm k\pi$, where $k = 0, 1, 2, 3$ and so on.